Lecture 3

Sylow theory

Objectives. *By Lagrange, the order of a subgroup divides $|G|$; by Cauchy, every prime divisor of $|G|$ is the order of an element (Lecture 1). This lecture asks which divisors of $|G|$ are orders of subgroups. Every prime power divisor is: Sylow's theorems give subgroups of the largest power of p, show they are conjugate, and count them. The count often forces a normal subgroup, and so decides that groups of many orders are not simple.*

By the end of this lecture you will be able to:

- Prove that a non-trivial p-group has a non-trivial centre, and that a group of order p^2 is abelian.
- State and prove Sylow's theorems, and list the possible numbers of Sylow p-subgroups from $|G|$ alone.
- Show that a group of order pq, for primes $p < q$, is $C_q \rtimes C_p$, cyclic when $p \nmid q-1$, and that groups of given orders are not simple.
- Find the Sylow subgroups of S_4 and A_4.

3.1 p-groups

Throughout, p is a prime. A *p-group* is a group of order p^a for some $a \geq 0$. Lagrange gives the first fact about them: every subgroup and every element has order a power of p.

Lemma 3.1. *If $G/Z(G)$ is cyclic, then G is abelian.*

Proof. Write $Z = Z(G)$, normal in G (§ 2.1), and let gZ generate G/Z. Each coset is $(gZ)^i = g^iZ$, so every element of G is g^iz with $z \in Z$. Two such elements commute, since z and z' commute with everything:

$$(g^iz)(g^jz') = g^{i+j}zz' = (g^jz')(g^iz).$$

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Theorem 3.2 (p-groups)**.** *Let P be a group of order p^a.*

(i) If $a \geq 1$, then $Z(P) \neq \mathbf{1}$.

(ii) If $a = 2$, then P is abelian, and $P \cong C_{p^2}$ or $P \cong C_p \times C_p$.

(iii) P has a normal subgroup of order p^b for each $0 \leq b \leq a$.

Proof. (i) Let P act on itself by conjugation. An element x is fixed by every g exactly when $gxg^{-1} = x$ for all g, so the fixed set is $Z(P)$. By Lemma 1.21,

$$|Z(P)| \equiv |P| \qquad \equiv 0 \pmod{p}.$$

As $e \in Z(P)$, the centre has at least p elements.

(ii) By the non-trivial-centre result, $|Z(P)|$ is p or p^2. If it were p, then $P/Z(P)$ would have order p, so it would be cyclic (Corollary 1.12), and P abelian by Lemma 3.1. But then $Z(P) = P$, of order p^2, not p. So $Z(P) = P$.

The two groups. If P has an element of order p^2, it is cyclic. If not, every $x \neq e$ has order p (Corollary 1.12).

In that case take $x \neq e$ and $y \notin \langle x \rangle$. The subgroup $\langle x \rangle \cap \langle y \rangle$ is a proper subgroup of $\langle y \rangle$, of order p, so it is $\mathbf{1}$.

Both are normal, as P is abelian, and $p \cdot p = |P|$. So, by Theorem 2.5(iii),

$$P \cong \langle x \rangle \times \langle y \rangle \cong C_p \times C_p.$$

(iii) By induction on a. For $b = 0$ take $\mathbf{1}$, so let $b \geq 1$; then $a \geq 1$.

A central subgroup of order p. The non-trivial centre and Lagrange show that p divides $|Z(P)|$, so by Cauchy (Theorem 1.22) there is $z \in Z(P)$ of order p. Then $N = \langle z \rangle$ has order p, and it is normal, since $gz^k g^{-1} = z^k$.

The induction. P/N has order p^{a-1}, so by induction it has a normal subgroup $\bar{K}$ of order p^{b-1}. By the correspondence theorem (Theorem 2.3(iii)), $\bar{K} = K/N$ for a normal subgroup $K \geq N$ of P, and

$$[P : K] = [P/N : \bar{K}] = p^{a-b}.$$

So $|K| = p^b$. ∎

The non-trivial-centre result fails without the prime power: $Z(S_3) = \mathbf{1}$ (Example 1.16). And (ii) stops at p^2: the p-group of the next example has order 8 and is not abelian.

Example 3.3 (A group of order 8)**.** Let $H_1 \leq S_4$ be the stabiliser of the pairing $\{12\,|\,34\}$ (§ 2.6):

$$H_1 = V_4 \cup \{(1\,2),\ (3\,4),\ (1\,3\,2\,4),\ (1\,4\,2\,3)\}.$$

It is not abelian: $(1\,2)$ and $(1\,3\,2\,4)$ do not commute. Its centre is

$$Z(H_1) = \{e,\ (1\,2)(3\,4)\},$$

where

$$(1\,2)(3\,4) = (1\,3\,2\,4)^2.$$

As the normal-subgroup theorem predicts, it has normal subgroups of each order 1, 2, 4, 8:

$$\mathbf{1} \trianglelefteq Z(H_1) \trianglelefteq V_4 \trianglelefteq H_1,$$

each normal in H_1; V_4 has index 2.

3.2 Existence

Write $|G| = p^a m$ with $p \nmid m$.

Definition 3.4 (Sylow subgroups)**.** A *Sylow p-subgroup* of G is a subgroup of order p^a. The set of them is $\mathrm{Syl}_p(G)$, and $n_p = \left|\mathrm{Syl}_p(G)\right|$.

So a Sylow p-subgroup is a p-subgroup as large as Lagrange allows. If $p \nmid |G|$, then $a = 0$ and $\mathbf{1}$ is the only one. Their existence rests on a congruence.

Lemma 3.5. *For $a \geq 0$ and $m \geq 1$,*

$$\binom{p^a m}{p^a} \equiv m \pmod{p}.$$

Proof. For $0 < k < p$, the prime p divides $p!$ but not $k!\,(p-k)!$, so it divides $\binom{p}{k}$. Hence, for polynomials with integer coefficients compared coefficient by coefficient modulo p,

$$(1+X)^p \equiv 1 + X^p.$$

Applying this a times, with $X^p, X^{p^2}, \ldots$ in place of X, and then taking the mth power,

$$(1+X)^{p^a} \equiv 1 + X^{p^a},$$
$$(1+X)^{p^a m} \equiv \left(1 + X^{p^a}\right)^m.$$

The coefficient of X^{p^a} is $\binom{p^a m}{p^a}$ on the left and $\binom{m}{1} = m$ on the right. ∎

Theorem 3.6 (Sylow: existence)**.** *G has a Sylow p-subgroup.*

Proof. Let Ω be the set of subsets $S \subseteq G$ with p^a elements. Left multiplication is injective, so $gS = \{gs : s \in S\}$ is again in Ω, and $g \cdot S = gS$ is an action of G on Ω.

An orbit of size prime to p. By Lemma 3.5,

$$|\Omega| = \binom{p^a m}{p^a} \equiv m \not\equiv 0 \pmod{p}.$$

The orbits partition Ω, so not all their sizes are multiples of p. Let S lie in an orbit with $p \nmid |G \cdot S|$, and let $H = G_S$.

$|H| \geq p^a$. By Corollary 1.11, $|G| = |G \cdot S|\ |H|$. So p^a divides $|G \cdot S|\ |H|$, and $p \nmid |G \cdot S|$; so p^a divides $|H|$.

$|H| \leq p^a$. Fix $s \in S$. For $h \in H$, $hs \in hS = S$; so $h \mapsto hs$ is an injective map $H \to S$, and

$$|H| \leq |S| = p^a.$$

So $|H| = p^a$, and H is a Sylow p-subgroup. ∎

Example 3.7 (Wielandt's count in S_3)**.** Take $p = 3$, so $p^a = 3$ and $m = 2$. There are $\binom{6}{3} = 20$ subsets of three elements, and $20 \equiv 2 \pmod 3$. The subset A_3 and its translates form one orbit, with stabiliser A_3:

$$\{A_3,\ (1\,2)A_3\}.$$

The orbit of size prime to 3 gives the Sylow 3-subgroup A_3.

Combined with Theorem 3.2(iii), existence answers the lecture's question for prime powers.

Corollary 3.8. *If p^b divides $|G|$, then G has a subgroup of order p^b.*

Proof. Then $b \leq a$. A Sylow p-subgroup has order p^a, and by Theorem 3.2(iii) it has a subgroup of order p^b. ∎

For divisors that are not prime powers the converse of Lagrange fails: A_4, of order 12, has no subgroup of order 6 (Example 3.19).

3.3 Conjugacy and counting

The Sylow p-subgroups are as tightly controlled as they are large.

Theorem 3.9 (Sylow: conjugacy and number)**.** *Let $|G| = p^a m$ with $p \nmid m$, and let $P \in \mathrm{Syl}_p(G)$.*

(i) Every p-subgroup Q of G lies in a conjugate of P: $Q \leq gPg^{-1}$ for some $g \in G$.

(ii) Any two Sylow p-subgroups are conjugate.

(iii) $n_p = [G : N_G(P)]$, and n_p divides m.

(iv) $n_p \equiv 1 \pmod{p}$.

Proof. (i) Let Q act on the left cosets G/P by $q \cdot gP = qgP$. There are $|G/P| = m$ cosets, so by Lemma 1.21 the number of fixed cosets is $\equiv m \not\equiv 0 \pmod{p}$. Let gP be fixed. For each $q \in Q$,

$$qgP = gP \iff g^{-1}qg \in P \qquad \iff q \in gPg^{-1}.$$

So $Q \leq gPg^{-1}$.

(ii) By containment, $Q \leq gPg^{-1}$ for a Sylow p-subgroup Q. Both have p^a elements; so $Q = gPg^{-1}$.

(iii) A conjugate of P has p^a elements, so by conjugacy the conjugates of P are exactly $\mathrm{Syl}_p(G)$. By Corollary 1.17 there are $[G : N_G(P)]$ of them. As $P \leq N_G(P) \leq G$, Lagrange gives

$$m = [G : P] = [G : N_G(P)]\,[N_G(P) : P].$$

(iv) Let P act on $\mathrm{Syl}_p(G)$ by conjugation; P itself is fixed. Let Q be fixed, that is, $P \leq N_G(Q)$, and write $N = N_G(Q)$.

P and Q are Sylow in N. Both lie in N and have p^a elements, the largest power of p dividing $|G|$, hence $|N|$.

$P = Q$. By Sylow conjugacy in N, $P = xQx^{-1}$ for some $x \in N$; and $xQx^{-1} = Q$, as x normalises Q.

So P is the only fixed point, and $n_p \equiv 1 \pmod{p}$ by Lemma 1.21. ∎

The divisibility and congruence bound n_p: it divides m and is $\equiv 1 \pmod{p}$.

Example 3.10 (Groups of order 24)**.** For $|G| = 2^3 \cdot 3$, n_3 divides 8 and n_2 divides 3:

$$n_3 \in \{1, 4\}, \qquad n_2 \in \{1, 3\}.$$

Corollary 3.11. *A Sylow p-subgroup P is normal in G if and only if $n_p = 1$.*

Proof. The normaliser formula gives $n_p = 1$ exactly when $N_G(P) = G$, which is normality (Definition 2.1). ∎

3.4 Order pq and non-simplicity

A proper normal subgroup $N \neq \mathbf{1}$ splits G into two smaller groups, N and G/N (Lecture 2); the groups with none are the building blocks.

Definition 3.12 (Simple group)**.** A group G is *simple* if $G \neq \mathbf{1}$ and its only normal subgroups are $\mathbf{1}$ and G.

A group of prime order is simple, by Lagrange. Lecture 4 proves that A_5 is simple. Sylow's theorems show that many orders admit no simple group.

Theorem 3.13 (Order pq)**.** *Let $p < q$ be primes and $|G| = pq$.*

(i) G has a normal subgroup $Q \cong C_q$, and $G = Q \rtimes P$ for any subgroup P of order p; such a P exists, and $P \cong C_p$.

(ii) If $p \nmid q - 1$, then $G \cong C_q \times C_p$, which is cyclic.

Proof. (i) By Theorem 3.9, n_q divides p and $n_q \equiv 1 \pmod q$. As $1 < p < q$, $p \not\equiv 1 \pmod q$; so $n_q = 1$. The Sylow q-subgroup Q is normal by Corollary 3.11, and cyclic by Corollary 1.12.

Let P be a subgroup of order p; one exists by Theorem 3.6, and it is cyclic. The order of $P \cap Q$ divides p and q, so $P \cap Q = \mathbf{1}$; and $|Q|\,|P| = |G|$. So $G = Q \rtimes P$ (Theorem 2.5).

(ii) Now n_p divides q and $n_p \equiv 1 \pmod p$. As $q \not\equiv 1 \pmod p$, $n_p = 1$, and P is normal too. By Theorem 2.5(iii),

$$G \cong Q \times P \cong C_q \times C_p,$$

which is cyclic as $\gcd(p, q) = 1$ (§ 2.3). ∎

Example 3.14 (Order 6)**.** Here $p = 2$ divides $q - 1 = 2$, and S_3 is not abelian. Its Sylow 3-subgroup A_3 is normal, its three Sylow 2-subgroups are the $\langle (i\,j) \rangle$, and $S_3 = A_3 \rtimes \langle (1\,2) \rangle$, with

$$(1\,2)(1\,2\,3)(1\,2)^{-1} = (1\,3\,2).$$

Thus the condition $p \nmid q - 1$ matters. By contrast $3 \nmid 5 - 1$, so every group of order 15 is cyclic.

When $p \mid q - 1$, a non-abelian group of order pq exists, and is unique up to isomorphism (Sheet 3). For $p = 2$ it is the dihedral group, as for $S_3 \cong D_3$ (Example 1.5).

Proposition 3.15 (Order $2q$)**.** *Let q be an odd prime. A group of order $2q$ is cyclic or isomorphic to D_q.*

Proof. By Theorem 3.13(i), $G = Q \rtimes P$ with $Q = \langle r \rangle$ of order q and $P = \langle s \rangle$ of order 2. As Q is normal, $srs^{-1} = r^k$ for some integer k.

The exponent. Conjugation by s is a homomorphism, so it carries r^k to $(r^k)^k$; and $s^2 = e$. So

$$r = s^2 r s^{-2} = s\, r^k\, s^{-1} = r^{k^2}.$$

So q divides $k^2 - 1 = (k - 1)(k + 1)$, and, q being prime, $k \equiv 1$ or $k \equiv -1 \pmod q$.

$k \equiv 1$. Then s commutes with r and with s, so it is central, and P is normal. As in the proof of Theorem 3.13(ii), $G \cong C_q \times C_2$, which is cyclic.

$k \equiv -1$. Then $srs^{-1} = r^{-1}$. Each element of G is $r^i s^j$ for exactly one $0 \le i < q$ and $j \in \{0, 1\}$ (Theorem 2.5(i)), and by Theorem 2.5(ii) the product is

$$(r^i s^j)(r^{i'} s^{j'}) = r^{\,i + (-1)^j i'}\, s^{\,j + j'}.$$

D_q, with its rotation r and reflection s, is also $\langle r \rangle \rtimes \langle s \rangle$ with $srs^{-1} = r^{-1}$ (§ 2.6), so its elements and its rule are the same, the exponents read modulo q and 2. Hence $r^i s^j \mapsto r^i s^j$ is an isomorphism $G \to D_q$. ∎

In D_q each of the q reflections $r^i s$ has order 2, so $n_2 = q$, the largest value Theorem 3.9 allows.

Corollary 3.16. *A group of order pq ($p < q$), or of order p^a with $a \ge 2$, is not simple.*

Proof. For pq, the subgroup Q of Theorem 3.13 is normal, of order neither 1 nor pq. For p^a, Theorem 3.2(iii) gives a normal subgroup of order p, neither 1 nor p^a. ∎

3.5 Worked example

What are the Sylow subgroups of S_4 and A_4, and what does Sylow theory say about every group of order 12?

The Sylow 3-subgroups of S_4. They have order 3, so each is generated by an element of order 3. These are the eight 3-cycles (§ 2.6), and each subgroup holds two of them. So $n_3 = 4$:

$$\langle (1\,2\,3)\rangle, \quad \langle (1\,2\,4)\rangle, \quad \langle (1\,3\,4)\rangle, \quad \langle (2\,3\,4)\rangle.$$

They are conjugate, as Theorem 3.9(ii) says; for instance

$$(3\,4)(1\,2\,3)(3\,4)^{-1} = (1\,2\,4).$$

The normaliser of $\langle (1\,2\,3)\rangle$ has index 4, so order 6. It contains the stabiliser S_3 of 4, which permutes $\{1,2,3\}$ and so carries $(1\,2\,3)$ to a 3-cycle on $\{1,2,3\}$, that is, to $(1\,2\,3)$ or $(1\,3\,2)$; so it is that S_3.

The Sylow 2-subgroups of S_4. They have order 8. The three stabilisers H_1, H_2, H_3 of the pairings

$$\{12\,|\,34\}, \qquad \{13\,|\,24\}, \qquad \{14\,|\,23\}$$

have order 8 (§ 2.6), and $n_2 \in \{1,3\}$, so $n_2 = 3$. They are conjugate: $(2\,3)$ carries $\{12\,|\,34\}$ to $\{13\,|\,24\}$, so

$$(2\,3)H_1(2\,3)^{-1} = H_2.$$

As $n_2 = [S_4 : N_{S_4}(H_1)]$, the normaliser of H_1 has order 8: it is H_1. By Theorem 3.9(i), each element of order 2 or 4 lies in some H_i; for instance $(1\,2\,3\,4)$ preserves $\{13\,|\,24\}$, so it lies in H_2.

The fixed-point count. H_1 acts on $\{H_1, H_2, H_3\}$ by conjugation. It fixes H_1; and $(1\,2) \in H_1$ exchanges $\{13\,|\,24\}$ and $\{14\,|\,23\}$, so it exchanges H_2 and H_3. The orbits are $\{H_1\}$ and $\{H_2, H_3\}$, and, as in the proof of Theorem 3.9(iv),

$$3 = 1 + 2 \qquad \equiv 1 \pmod 2.$$

A_4. Its order is $12 = 2^2 \cdot 3$. The 3-cycles are even, so its Sylow 3-subgroups are the four above, and $n_3 = 4$. Its Sylow 2-subgroups have order 4, and $V_4 \trianglelefteq A_4$ is one; so $n_2 = 1$ by Corollary 3.11. With S_3 (Example 3.14):

G	$\lvert G\rvert$	n_2	n_3
S_3	$2 \cdot 3$	3: the $\langle (i\,j)\rangle$	1: A_3
A_4	$2^2 \cdot 3$	1: V_4	4: the $\langle (i\,j\,k)\rangle$
S_4	$2^3 \cdot 3$	3: H_1, H_2, H_3	4: the $\langle (i\,j\,k)\rangle$

Proposition 3.17. *No group of order 12 is simple.*

Proof. Let $|G| = 12 = 2^2 \cdot 3$. By Theorem 3.9, $n_3 \in \{1,4\}$. If $n_3 = 1$, the Sylow 3-subgroup is normal (Corollary 3.11), of order 3, neither 1 nor 12.

Let $n_3 = 4$. Two distinct subgroups of order 3 meet in a subgroup of order dividing 3, not 3; so in **1**. The four of them hold $4 \cdot 2 = 8$ elements of order 3. A Sylow 2-subgroup has 4 elements, none of order 3, so it lies among the other

$$12 - 8 = 4$$

elements, and is equal to them. So $n_2 = 1$, and the Sylow 2-subgroup is normal, of order 4. ∎

A_4 is the second case: $n_3 = 4$ forces $n_2 = 1$, and there V_4 is the normal subgroup.

Example 3.18 (Guard: Sylow subgroups can overlap)**.** The count used that distinct subgroups of prime order meet in $\mathbf{1}$. Sylow subgroups of order p^a with $a \geq 2$ can share more: in S_4,

$$H_1 \cap H_2 = V_4,$$

so the three Sylow 2-subgroups hold 16 elements. Counting their non-identity elements separately would incorrectly give

$$3 \cdot 7 + 1 = 22.$$

Counting elements subgroup by subgroup needs the subgroups to meet in $\mathbf{1}$; prime order guarantees it, as for the orders 30 and 56 of Sheet 3.

Example 3.19 (Guard: the converse of Lagrange fails)**.** 6 divides $|A_4| = 12$, but A_4 has no subgroup of order 6. First, the classes of A_4:

- $(1\,2)(3\,4)$. Its class lies in its S_4-class, the three products of two transpositions, and it contains all three:
$$(1\,2\,3)\,(1\,2)(3\,4)\,(1\,2\,3)^{-1} = (1\,4)(2\,3),$$
$$(1\,3\,2)\,(1\,2)(3\,4)\,(1\,3\,2)^{-1} = (1\,3)(2\,4).$$
So its class has 3 elements.
- *The* 3*-cycles.* $C_{A_4}((1\,2\,3))$ lies in $C_{S_4}((1\,2\,3))$, of order $z_{(3,1)} = 3$ (Theorem 2.9), and contains $\langle(1\,2\,3)\rangle$; so it is $\langle(1\,2\,3)\rangle$, and the class of $(1\,2\,3)$ has $12/3 = 4$ elements (Theorem 1.15). The same holds for every 3-cycle, so the eight 3-cycles fall into two classes of 4.

So the class sizes are 1, 3, 4, 4. A subgroup of order 6 would have index 2, so it would be normal (Proposition 2.2(iii)), so a union of classes containing e (Proposition 2.2(ii)). The possible sizes are

$$\{1, 4, 5, 8, 9, 12\},$$

and 6 is not among them. So Corollary 3.8 needs the prime power.

Self-check. Is every group of order 45 abelian? *Answer:* yes. As $45 = 3^2 \cdot 5$, n_5 divides 9 and is $\equiv 1 \pmod 5$, and n_3 divides 5 and is $\equiv 1 \pmod 3$; so $n_5 = n_3 = 1$.

The Sylow subgroups P_3, of order 9, and P_5, of order 5, are then normal and meet in $\mathbf{1}$, so by Theorem 2.5(iii)

$$G \cong P_3 \times P_5.$$

P_3 is abelian by Theorem 3.2(ii), and P_5 is cyclic; so G is abelian.

Summary.

- *The main results:* Sylow p-subgroups exist, are conjugate, and number $n_p \equiv 1 \pmod p$, a divisor of the prime-to-p factor m in $|G| = p^a m$ (Theorem 3.6, Theorem 3.9); every prime power dividing $|G|$ is the order of a subgroup (Corollary 3.8), but not every divisor (Example 3.19).
- *Handed on:* a non-trivial p-group has a non-trivial centre (Theorem 3.2); simple groups, and the orders pq, p^a $(a \geq 2)$ and 12 that have none (Corollary 3.16, Proposition 3.17).
- *Next:* Lecture 4 proves A_5 simple, and builds finite groups from simple pieces; p-groups are its first nilpotent groups.

References

Humphreys, *A Course in Group Theory*: p-groups, Props. 10.20–10.21 and Cor. 10.22, pp. 94–95; the binomial congruence, Lemma 11.2, p. 98; existence by Wielandt's argument, Thm 11.4, p. 99; conjugacy and number, Thms 11.6 and 11.10, Cor. 11.11, pp. 100–102; order 12, Prop. 12.3, p. 108. Dummit and Foote, *Abstract Algebra*: simple groups, §3.4, p. 102; p-groups, §4.3, Thm 8 and Cor. 9, p. 125; Sylow's theorems, §4.5, Thm 18 and Cor. 20, pp. 139–142, with S_4, order pq and order 12, pp. 142–144; normal subgroups of every order p^b, §6.1, Thm 1, p. 188. Isaacs, *Finite Group Theory*: the congruence and Wielandt's proof, Lemma 1.8 and Thm 1.7, pp. 9–10.